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Pontryagin's maximum principle for dynamic systems on time scales

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ABSTRACT

In this work, an analogue of Pontryagin's maximum principle for dynamic equations on time scales is given, combining the continuous and the discrete Pontryagin maximum principles and extending them to other cases 'in between'. We generalize known results to the case when a certain set of admissible values of the control is not necessarily closed (but convex) and the attainable set is not necessarily convex. At the same time, we impose an additional condition on the graininess of the time scale. For linear systems, sufficient conditions in the form of the maximum principle are obtained.

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1. Introduction

Optimal control is one of the most important areas of extreme value theory. It has great practical applications in various fields of human activity. The main research methods of optimal control problems are the method of Bellman dynamic programming [12] and the Pontryagin maximum principle (denoted in short by PMP) [51].

Since the release of these pioneering works, there appeared many generalizations on the different classes of dynamical systems, such as finite and infinite, deterministic and stochastic systems [2,27,28,31–33,37,47].

Following this discovery, the corresponding theory has been developed for discrete-time dynamics, under appropriate convexity assumptions [26,38,44,45], leading to a version of the PMP for discrete-time optimal control problems. The dynamic programming method for discrete systems was considered, for example, in [48,53]. The first motivation of the study of such problems comes from economic theory, notably from optimal growth theory [48,50,53]. It can be noted that many counter-examples were provided in [26], showing that, as is now well known, the exact analogue of the continuous-time PMP does not hold at the discrete level. More precisely, the maximization condition of the PMP cannot be expected to hold in general in the discrete-time case. Nevertheless a weaker condition can be derived, see [26, Theorem 42.1 on p. 330].

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Time scales theory was introduced in [40] in order to unify discrete and continuous analysis. A time scale $\mathbb{T}$ is an arbitrary nonempty closed subset of $\mathbb{R}$, and a dynamic system is said to be posed on the time scale $\mathbb{T}$ whenever the time variable evolves along this set $\mathbb{T}$. The continuous-time case corresponds to $\mathbb{T} = \mathbb{R}$, and the discrete-time case corresponds to $\mathbb{T} = \mathbb{Z}$. Because unification and extension, the two main features of the calculus on time scales, are of significance in mathematical modelling, there exist many potential applications for this relatively new theory [4,5,15,16,18,19].

In recent years, many authors studied the problem of optimal control on time scales, see, e.g., [1,7,8,10,22–25,34,39,49,52]. The theory of calculus of variations on time scales, initiated in [13], has been well studied in the existing literature (see, e.g., [9,11,14,17,35,36, 42]).

With regard to the method of dynamic programming by Bellman, we note the paper [55], where the Hamilton–Jacobi–Bellman one-dimensional equations for optimal control of the equation that contains a linear part are obtained, and the paper [46], where the general case is considered. Here, necessary and sufficient conditions for the linear quadratic regulator to dynamic equations on time scales are obtained.

Few attempts have been made to derive a PMP on time scales. In [41], the authors establish a weak PMP for shifted controlled systems, a strong version of the PMP is claimed in [54]. As stated in [29], many arguments therein are incomplete.

Our work is based essentially on the results of [29]. In [29], in order to prove the PMP, the authors use Ekeland's variational principle. For this, it is necessary to impose the following conditions on the initial optimal control problem:

(1) The set Ω of values of admissible controls is closed;
(2) The set S of restrictions on the admissible trajectories is convex.

Of course, these conditions are not satisfied for all optimal control problems. For example, for the restriction $x(t_1) \leq x^2(t_0)$, the set S is not convex. The purpose of this work is to remove these restrictions. To achieve this goal, we will use an approach different from [29] based on 'a package of needles'. This will reduce the initial problem to a finite-dimensional smooth (in a cone) optimization problem with equalities and inequalities constraints, with further application of the method of Lagrange multipliers.

For the sake of fairness, it must be noted that while removing the mentioned restrictions (1) and (2) of [29], we introduce some other conditions. Here, the set of admissible controls Ω (in our work denoted by U) will be assumed to be convex. However, this condition is imposed for simplicity and ease of presentation, and it can be replaced by a condition such as dense-directional convexity, as in [29]. More important are the conditions imposed on the graininess of the time scale. As will be shown, these conditions are not necessarily satisfied for all time scales. However, they give a strong maximum principle at right-scattered points which are left-dense at the same time.

This paper is organized as follows. In Section 2, we introduce some preliminary definitions and results about the calculus on time scales (Section 2.1) and we give rigorous formulations of the considered problems and state two theorems (Section 2.2). The first theorem provides necessary conditions for optimality (PMP), the second one provides sufficient conditions of optimality for the linear-convex case. Section 3 is devoted to the proofs of the main results. In Section 3.1, we build 'a package of needles' consisting of variations of optimal control and we present some statements about the built package. In

Section 3.2, for completeness, we recall necessary optimality conditions for a smooth (in a cone) extremal problem with limitations such as equalities and inequalities. In Section 3.3, we present results on the relationship of the fundamental matrices of solutions for linear conjugated systems. The proofs of Theorems 2.11 and 2.13 are given in Sections 3.4 and 3.5, respectively.

2. Problem statements and main results

2.1. Preliminaries on time scales

Let $\mathbb{T}$ be a time scale, that is, an arbitrary closed nonempty subset of $\mathbb{R}$. The distance between the points $t, s \in \mathbb{T}$ is defined as the normal distance on $\mathbb{R}^1$: $|t - s|$. We assume that $\mathrm{card}(\mathbb{T}) \geq 2$. For every $A \subset \mathbb{R}$, we denote $A_{\mathbb{T}} = A \cap \mathbb{T}$. We define the forward and backward jump operators $\rho, \sigma : \mathbb{T} \to \mathbb{T}$ by

$$\sigma(t) = \inf\{s \in \mathbb{T} : \ s > t\} \quad \text{and} \quad \rho(t) = \sup\{s \in \mathbb{T} : \ s < t\}$$

(supplemented by $\inf \emptyset := \sup \mathbb{T}$ and $\sup \emptyset := \inf \mathbb{T}$). The graininess function $\mu : \mathbb{T} \to [0, \infty)$ is defined by

$$\mu(t) = \sigma(t) - t.$$

A point $t \in \mathbb{T}$ is called *left-dense* (*left-scattered*, *right-dense*, *right-scattered*) if $\rho(t) = t$ ($\rho(t) < t, \sigma(t) = t, \sigma(t) > t$) hold. If $\mathbb{T}$ has a left-scattered maximum M, then we define $\mathbb{T}^\kappa = \mathbb{T} \setminus \{M\}$; otherwise, we set $\mathbb{T}^\kappa = \mathbb{T}$. For a function $f : \mathbb{T} \to \mathbb{R}$, we denote $f^\sigma = f \circ \sigma$.

Definition 2.1: A function $f : \mathbb{T} \to \mathbb{R}$ is called rd-continuous if it is continuous at each right-dense point $t \in \mathbb{T}$ and left-hand limits exist at each left-dense point $t \in \mathbb{T}$.

Definition 2.2: A function $f : [a, b]_{\mathbb{T}} \to \mathbb{R}$ is called piecewise continuous on $[a, b]_{\mathbb{T}}$ if it is continuous at all points in $[a, b]_{\mathbb{T}}$ except for a finite number of points at which it is rd-continuous.

Note that at an isolated (i.e., left-scattered and right-scattered) point, any function is considered to be continuous.

Definition 2.3: A function $f : [a, b]_{\mathbb{T}} \to \mathbb{R}$ is Δ-differentiable at $t \in \mathbb{T}^\kappa$ provided there exists an $\alpha \in \mathbb{R}$ such that for each $\varepsilon > 0$, there exists a neighborhood B of t such that

$$|f(\sigma(t)) - f(s) - \alpha(\sigma(t) - s)| \leq \varepsilon |\sigma(t) - s| \quad \text{for all} \quad s \in B \cap \mathbb{T}.$$

We denote α by $f^\Delta(t)$.

We denote by RS the set of all right-scattered points of $\mathbb{T}$, and by RD the set of all right-dense points of $\mathbb{T}$ in $\mathbb{T} \setminus \{\sup \mathbb{T}\}$. Similarly, we denote by LS (LD) the set of all left-scattered (left-dense) points of $\mathbb{T}$.

For every $a, b \in \mathbb{T}$, $a \neq b$, and every $s \in [a, b)_{\mathbb{T}} \cap \mathrm{RD}$, we set

$$\mathcal{V}_s^b = \{\beta \geq 0 : \ s + \beta \in [s, b)_{\mathbb{T}}\},$$

and for every $s \in (a, b]_{\mathbb{T}} \cap \mathrm{LD}$, we set

$$\mathcal{V}_a^s = \{\beta \geq 0 : \ s - \beta \in (a, s]_{\mathbb{T}}\}.$$

Let $t_0, t_1 \in \mathbb{T}$ such that $t_0 < t_1$ and $t_1 \in \mathbb{T}^\kappa$. In this paper, an important rôle is played by the following condition on the time scale.

Definition 2.4: We say that at the point $s \in [t_0, t_1)_{\mathbb{T}} \cap \mathrm{RD}$, the condition of *density* is satisfied if

$$\lim_{\substack{\beta \to 0 \\ \beta \in \mathcal{V}_s^{t_1}}} \frac{\mu(s+\beta)}{\beta} = 0. \tag{2.1}$$

Similarly, at the point $s \in (t_0, t_1]_{\mathbb{T}} \cap \mathrm{LD}$, the condition of *density* is satisfied if

$$\lim_{\substack{\beta \to 0 \\ \beta \in \mathcal{V}_{t_0}^{s}}} \frac{\mu(s-\beta)}{\beta} = 0. \tag{2.2}$$

Example 2.5: Note that conditions (2.1) and (2.2) are not necessarily satisfied for all time scales. Obviously, if $\mathbb{T} = \mathbb{R}$, then (2.1) and (2.2) are satisfied. From the properties of the Cantor set, it follows that if $\mathbb{T}$ is the Cantor set, then (2.1) and (2.2) are also fulfilled. For the time scale

$$\mathbb{T} = \left\{ \frac{1}{n} : \ n \in \mathbb{N} \right\} \cup \{0\},$$

it is not difficult to see that (2.1) is satisfied. However, for the time scale

$$\mathbb{T} = \{2^{-n} : \ n \in \mathbb{N}\} \cup \{0\},$$

(2.1) is not satisfied.

2.2. Problem statements and main results

By $(\cdot, \cdot)$, we denote the scalar product in $\mathbb{R}^d$, and by $|\cdot| = \sqrt{(\cdot,\cdot)}$ the Euclidean norm in $\mathbb{R}^d$. The matrix norm is denoted by $\|\cdot\|$. Other norms will be explained as they appear.

Let $t_0, t_1 \in \mathbb{T}$ such that $t_0 < t_1$ and $t_1 \in \mathbb{T}^\kappa$. We consider the following optimal control problem on the interval $[t_0, t_1]_{\mathbb{T}}$:

$$\Phi_0(x(t_0), x(t_1)) \to \inf, \tag{2.3}$$

$$x^\Delta(t) = f(t, x(t), u(t)), \quad t \in [t_0, t_1]_{\mathbb{T}}, \tag{2.4}$$

$$\begin{aligned} &\Phi_i(x(t_0), x(t_1)) = 0, \quad i \in \{1, \dots, k\}, \\ &\Phi_i(x(t_0), x(t_1)) \le 0, \quad i \in \{k+1, \dots, n\}. \end{aligned} \tag{2.5}$$

Here, $x \in D$ is the state, D is a domain in $\mathbb{R}^d$, $u \in U \subset \mathbb{R}^m$, and U is a convex set. The functions f, f'_x, f'_u are defined on $t \in [t_0, t_1]_{\mathbb{T}}$, $x \in D$, and $u \in U$ are continuous in all arguments. The functions Φ_i, $i \in \{0, \dots, n\}$, are defined in some domain $W \subset \mathbb{R}^d \times \mathbb{R}^d$ and are smooth in their arguments. We assume that the controls $u : [t_0, t_1]_{\mathbb{T}} \to U$ are piecewise continuous functions.

Definition 2.6: A pair (x, u) is called a controlled process in the problem (2.3)–(2.5) if

(1) the control $u : [t_0, t_1]_{\mathbb{T}} \to U$ is piecewise continuous;
(2) the state trajectory $x : [t_0, t_1]_{\mathbb{T}} \to D$ is continuous;

(3) for all $t \in [t_0, t_1]_{\mathbb{T}}$ except, possibly, at the points of discontinuity of the control u, the function x satisfies the differential equation

$$x^{\Delta}(t) = f(t, x(t), u(t)),$$

i.e., at all points $t \in [t_0, t_1]_{\mathbb{T}}$, the state trajectory x satisfies the integral equation

$$x(t) = x(t_0) + \int_{[t_0,t)_{\mathbb{T}}} f(s, x(s), u(s))\Delta s. \tag{2.6}$$

In the case of Definition 2.6, we say that x corresponds to the control u. If, besides this, (2.5) is satisfied, then the pair (x, u) is called an admissible controlled process.

Remark 2.7: It is possible to consider a wider and more typical space of controls – the space $L^{\infty}([t_0, t_1]_{\mathbb{T}}, U)$ – in Definition 2.6 and adapt our proofs to this class. Here, condition (4)(b) (see below) from Theorem 2.11 will be true Δ-a.e. for $s \in [t_0, t_1)_{\mathbb{T}} \cap \mathrm{RD}$. Our choice of the space of piecewise continuous controls is motivated by technically complicated proofs. Due to this choice, we simplify them a little bit. It is especially true for the proof of Lemma 3.2.

Definition 2.8: A controlled admissible process $(\hat{x}, \hat{u})$ is called (locally) optimal if there exists $\varepsilon > 0$ such that for any admissible controlled process (x, u) with

$$|x(t) - \hat{x}(t)| < \varepsilon \quad \text{for each} \quad t \in [t_0, t_1]_{\mathbb{T}}, \tag{2.7}$$

the inequality

$$\Phi_0(x(t_0), x(t_1)) \geq \Phi_0(\hat{x}(t_0), \hat{x}(t_1)) \tag{2.8}$$

holds.

The main results of this paper are the following two theorems. The first theorem is concerned with necessary conditions for optimality in problem (2.3)–(2.5) (PMP), and the second theorem provides sufficient optimality conditions for a linear system.

Let $(\hat{x}, \hat{u})$ be an optimal process for problem (2.3)–(2.5) in the sense of Definition 2.8.

Definition 2.9: The linear system of differential equations

$$\Psi^{\Delta} = -\left(\frac{\partial f(t, \hat{x}(t), \hat{u}(t))}{\partial x}\right)^{*} \Psi^{\sigma} \tag{2.9}$$

is called conjugate to the system (2.4). Here, $*$ denotes transposition.

Definition 2.10: The function

$$H(t, \Psi, x, v) = (\Psi, f(t, x, v)) = \sum_{i=1}^{d} \left(\Psi_i, f_i(t, x, v)\right)$$

is called a Pontryagin function, and the function

$$L\left(\lambda, x(t_0), x(t_1)\right) = \sum_{i=0}^{n} \lambda_i \Phi_i(x(t_0), x(t_1))$$

is called a Lagrange function of the problem (2.3)–(2.5). Here, $\lambda_0, \lambda_1, \dots, \lambda_n$ are called Lagrange multipliers.

Theorem 2.11 (Pontryagin's Maximum Principle): *Suppose that for functions f and Φ_i, $i \in \{0, \dots, n\}$, the conditions of problem (2.3)–(2.5) are satisfied and for points $s \in [t_0, t_1)_{\mathbb{T}} \cap \mathrm{RD}$ and $s \in (t_0, t_1]_{\mathbb{T}} \cap \mathrm{LD}$, the conditions of density (2.1) and (2.2) are satisfied, respectively. If $(\hat{x}, \hat{u})$ is an optimal process for (2.3)–(2.5), then there exists a nonzero vector $\lambda = (\lambda_0, \lambda_1, \dots, \lambda_n)$ – the Lagrange multipliers – and $\Psi(t) = \big(\Psi_1(t), \dots, \Psi_d(t)\big)^*$ – a solution of the conjugate system – such that the following conditions hold:*

(1) *nonnegativity:*

$$\lambda_i \geq 0 \quad \text{for} \quad i = 0 \quad \text{and} \quad i \in \{k+1, \dots, n\};$$

(2) *complementary slackness:*

$$\lambda_i \Phi_i(\hat{x}(t_0), \hat{x}(t_1)) = 0 \quad \text{for all} \quad i \in \{k+1, \dots, n\};$$

(3) *transversality conditions:*

$$\Psi(t_0) = L_{x_0} := \frac{\partial L}{\partial x(t_0)} \quad \text{and} \quad -\Psi(t_1) = L_{x_1} := \frac{\partial L}{\partial x(t_1)};$$

(4) *maximum condition:*

(a) *weaker maximum condition: for every point $r \in [t_0, t_1)_{\mathbb{T}} \cap \mathrm{RS}$,*

$$\left(\frac{\partial H}{\partial u} \left(r, \Psi^{\sigma}(r), \hat{x}(r), \hat{u}(r)\right), v - \hat{u}(r) \right)_{\mathbb{R}^n} \leq 0 \quad \text{for all} \quad v \in U. \tag{2.10}$$

Note that (2.10) is satisfied for $r = t_0$ if $t_0 \in \mathrm{RS}$;

(b) *stronger maximum condition: if $s \in [t_0, t_1)_{\mathbb{T}} \cap \mathrm{RD}$, then*

$$\max_{v \in U} H\left(s, \Psi(s), \hat{x}(s), v\right) = H\left(s, \Psi(s), \hat{x}(s), \hat{u}(s)\right). \tag{2.11}$$

Equality (2.11) also holds for $s \in (t_0, t_1]_{\mathbb{T}} \cap \mathrm{LD}$ provided $\hat{u}$ is continuous at s.

We also note that the formulation of Theorem 2.11 is not restricted to closedness of the set of admissible controls U and is not restricted to convexity of the set of restrictions defined in $\mathbb{R}^d \times \mathbb{R}^d$ by (2.5).

Let us introduce an example, illustrating Theorem 2.11.

Example 2.12: Let $\mathbb{T} = [0, 3] \cup [4, \infty)$ and $t_0 = 0$, $t_1 = 4$, i.e., $[t_0, t_1]_{\mathbb{T}} = [0, 3] \cup \{4\}$. Let us consider the optimal control problem

$$J = \int_{[0,4)_{\mathbb{T}}} (u^2(t) + x(t)) \Delta t \to \inf, \quad |u| \leq 1, \quad x(4) = 0.$$

If we introduce an additional coordinate $x_0(t) = \int_{[0,t)_{\mathbb{T}}} (u^2(s) + x(s))\Delta s$, then the problem is written as (2.3)–(2.5)

$$\begin{cases} x_0(4) \to \inf \\ x_0^\Delta = x + u^2, \quad x^\Delta = u, \quad x(4) = 0, \quad x_0(0) = 0, \quad |u| \le 1. \end{cases}$$

In this case, the adjoint system (2.9) has the form

$$\begin{cases} \Psi_0^\Delta = 0 \\ \Psi_1^\Delta = -\Psi_0^\sigma. \end{cases}$$

Hence, $\Psi_0 = C_0$, $\Psi_1 = -C_0 t + C_1$. The Lagrange function is $L = \lambda_0 x_0(4) + \lambda_1 x(4) + \lambda_2 x_0(0)$. The transversality conditions give the equalities $C_0 = \lambda_2 = -\lambda_0$, $C_1 = 0$, $4C_0 = \lambda_1$. If $\lambda_0 = 0$, then $\lambda_2 = 0$ and $C_0 = 0$, therefore $\lambda_1 = 0$, which is impossible. Consequently, $\lambda_0 \neq 0$ and $\lambda_0 = 1$. It means that $\Psi_0 = -1$, $\Psi_1 = t$. The Pontryagin function is $H = -(x + u^2) + ut$. It is easily seen that the stronger maximum condition (2.11) gives the equalities

$$\hat{u}(t) = \begin{cases} \frac{t}{2}; & t \in [0, 2) \\ 1; & t \in [2, 3) \cup \{4\}. \end{cases}$$

The weaker maximum condition leads to the inequality $(-2\hat{u}(3) + 3)(v - \hat{u}(3)) \le 0$, that is equivalent to

$$\begin{cases} \hat{u}(3) \le \frac{3}{2} \\ \hat{u}(3) \ge v \end{cases}$$

for any v such that $|v| \le 1$. Hence, $\hat{u}(3) = 1$. It is not hard to prove, using Definition 2.8, that the control

$$\hat{u}(t) = \begin{cases} \frac{t}{2}; & t \in [0, 2) \\ 1; & t \in [2, 3] \cup \{4\}, \end{cases}$$

is optimal indeed.

We now consider the linear-convex optimal control problem on $[t_0, t_1]_{\mathbb{T}}$:

$$\Phi_0(x(t_0), x(t_1)) \to \inf, \tag{2.12}$$

$$x^\Delta = A(t)x + B(t)u(t), \quad t \in [t_0, t_1]_{\mathbb{T}}, \tag{2.13}$$

$$\begin{aligned} \Phi_i(x(t_0), x(t_1)) &= 0, \quad i \in \{1, \dots, k\}, \\ \Phi_i(x(t_0), x(t_1)) &\le 0, \quad i \in \{k+1, \dots, n\}. \end{aligned} \tag{2.14}$$

Here, $u \in U$, $U \subset \mathbb{R}^m$ is convex, the matrix-valued functions A, B are continuous, and Φ_i are convex and smooth functions in $\mathbb{R}^d \times \mathbb{R}^d$.

Theorem 2.13: *In order to make an admissible process $(\hat{x}, \hat{u})$ optimal in (2.12)–(2.14), a sufficient condition is the existence of Lagrange multipliers $\lambda_0, \lambda_1, \dots, \lambda_n$ and solutions $\Psi = (\Psi_1, \dots, \Psi_d)^*$ of the conjugate system (2.9) such that the following conditions are satisfied:*

(1) *nonnegativity:*

$$\lambda_0 > 0 \quad \textit{and} \quad \lambda_i \ge 0 \quad \textit{for all} \quad i \in \{k+1, \dots, n\};$$

(2) *complementary slackness:*

$$\lambda_i \Phi_i(\hat{x}(t_0), \hat{x}(t_1)) = 0 \quad \textit{for all} \quad i \in \{k+1, \dots, n\};$$

(3) *transversality conditions:*

$$\Psi(t_0) = L_{x_0} \quad \textit{and} \quad \Psi(t_1) = -L_{x_1},$$

where

$$L = \sum_{i=1}^{n} \lambda_i \Phi_i$$

is the Lagrange function;

(4) *maximum condition:*

$$\max_{v \in U} \left(\Psi^{\sigma}(t), B(t)v\right) = \left(\Psi^{\sigma}(t), B(t)\hat{u}(t)\right) \quad \textit{for all} \quad t \in [t_0, t_1]_{\mathbb{T}}.$$

Remark 2.14: It is easy to see that for system (2.13), the weak maximum condition (2.10) becomes the stronger condition (2.11).

3. Proofs of the main results

3.1. The package of needles

In order to prove Theorem 2.11, we include the optimal process $(\hat{x}, \hat{u})$ in a special family of admissible processes – the 'package' of needle-like variations. Further calculations we carry out for right-scattered and right-dense points. For left-dense points, considerations are similar.

The desired family of controlled processes depends on the following parameters:

(1) initial data $x(t_0) = x_0$;
(2) $\bar{\alpha} := (\alpha_1, \dots, \alpha_N)$, where α_i are sufficiently small and $\alpha := \sum_{i=1}^{N} \alpha_i$;
(3) $\bar{\tau} := (\tau_1, \dots, \tau_N)$, where $\tau_i \in (t_0, t_1)_{\mathbb{T}} \cap \mathrm{RD}$, $\tau_1 \le \tau_2 \le \dots \le \tau_N$;
(4) $\bar{v} := (v_1, \dots, v_N)$, where $v_i \in U$;
(5) $\bar{r} := (r_1, \dots, r_\nu)$, where $r_i \in (t_0, t_1)_{\mathbb{T}} \cap \mathrm{RS}$, $r_1 \le r_2 \le \dots \le r_\nu$;
(6) $\bar{z} := (z_1, z_2, \dots, z_\nu)$, where $z_i \in U$;
(7) $\bar{\gamma} := (\gamma_1, \dots, \gamma_\nu)$, where $\gamma_i \ge 0$ are sufficiently small.

Subsequently, $\bar{\tau}, \bar{v}, \bar{r}, \bar{z}$ are considered to be fixed and $x_0, \bar{\alpha}, \bar{\gamma}$ are variables, and various functions will be differentiated with respect to them.

The package of needle-like variations, which we want to make up, will consist of two types of needle-like variations: variations by right-dense points and variations by right-scattered points. When $N = 1$ and $\nu = 0$, such package consists of one needle-like variation for right-dense points and coincides with the needle-like variation $II = (s, z)$ from [29, Section 3.2.3]. In our case, $II = (\tau_i, v_i)$. If $N = 0$ and $\nu = 1$, such package consists of one needle-like variation for right-scattered points and coincides with the needle-like variation $\Pi = (r, y)$ from [29, Section 3.2]. In our case, $\Pi = (r_i, z_i)$.

However, the structure of our package is more difficult. It unifies any finite number of elementary needles both of the first and second kind. It is significant that some of these

needles may differ only by the value of controls v_i or z_i at the same τ_i or r_i. Therefore, defining the needles by formulas from [29] is no longer admissible: for the same τ_i or r_i, different needles will overlap. To avoid this, we proceed as follows.

Suppose that among the needles of the first type, there are duplicate sets of needles

$$(\tau_i, v_{1,i}), \quad (\tau_i, v_{2,i}), \quad \cdots, \quad (\tau_i, v_{q_i,i}). \tag{3.1}$$

Here, q_i is the number of repetitions of needles with the same τ_i. In the package, there may be several such sets. For each of them, we define the nonoverlapping intervals

$$\Delta_{s,i} = [\tau_i + (s-1)\bar{\alpha}_i, \tau_i + (s-1)\bar{\alpha}_i + \alpha_{s,i})_{\mathbb{T}},$$

where $s \in \{1, \dots, q_i\}$, $\alpha_{s,i} \geq 0$ are sufficiently small, and $\bar{\alpha}_i = \sum_{s=1}^{q_i} \alpha_{s,i}$.

Let us assign the control $v_{s,i}$ of the sequence (3.1) to each interval $\Delta_{s,i}$. As a result, instead of intersecting q_i intervals (3.1), we obtain q_i disjoint intervals $\Delta_{s,i}$; at each of them, there is the admissible control $v_{s,i}$. We denote

$$\Delta_i = \bigcup_{s=1}^{q_i} \Delta_{s,i}.$$

Some of the Δ_i may be the union of several intervals and some may be an interval (depending on how many needle-like variations there are in the package with the same τ_i). Since there are N different pairs of basic needles (τ_i, v_i), there are also N intervals $\Delta_{s,i}$.

Now, for convenience, let us renumber all $\alpha_{s,i}$ and change their notation to α_i, $i \in \{1, \dots, N\}$. As a result, for the needle-like variations of the first kind, we obtain the following variation of the optimal control $\hat{u}$:

$$u(t, \bar{\alpha}, \bar{\tau}, \bar{v}) = \begin{cases} \hat{u}(t) & \text{if } t \in [t_0, t_1]_{\mathbb{T}} \setminus \bigcup_{i=1}^{N} \Delta_i, \\ v_{s,i} & \text{if } t \in \Delta_{s,i},\ s \in \{1, \dots, q_i\},\ i \in \{1, \dots, N\}. \end{cases} \tag{3.2}$$

Let us turn to the second type of needles. If some of them are repetitive needles $(r_i, z_1), (r_i, z_2), \dots, (r_i, z_{l_i})$, then the corresponding variation is as follows:

$$u(t, \bar{\gamma}, \bar{r}, \bar{z}) = \begin{cases} \hat{u}(r_i) + \gamma_{1,i}(z_1 - \hat{u}(r_i)) + \cdots + \gamma_{l_i,i}(z_{l_i} - \hat{u}(r_i)) \\ \qquad \text{if } t = r_i,\ i \in \{1, \dots, \nu\}, \\ \hat{u}(t) \quad \text{if } t \neq r_i,\ i \in \{1, \dots, \nu\}. \end{cases} \tag{3.3}$$

Here, $\gamma_{p,i} \geq 0$, $p \in \{1, \dots, l_i\}$, l_i is the number of repetitions of the needles with the same r_i in the package, $i \in \{1, \dots, \nu\}$. Now, let us unify packages (3.2) and (3.3) into one package

and construct the following variation of the optimal control $\hat{u}$:

$$u(t,\bar{\alpha},\bar{\tau},\bar{\nu},\bar{\gamma},\bar{r},\bar{z}) = \begin{cases} \hat{u}(t) & \text{if } t \in [t_0,t_1]_{\mathbb{T}} \setminus \bigcup_{i=1}^{N} \Delta_i \setminus \bigcup_{i=1}^{\nu}\{r_i\}, \\ v_{s,i} & \text{if } t \in \Delta_{s,i},\ s \in \{1,\dots,q_i\},\ i \in \{1,\dots,N\}, \\ \hat{u}(r_i) + \displaystyle\sum_{p=1}^{l_i} \gamma_{p,i}\left(z_p - \hat{u}(r_i)\right) & \\ & \text{if } t = r_i,\ i \in \{1,\dots,\nu\}. \end{cases} \tag{3.4}$$

At the same time, we choose $\alpha_i > 0$ so small that the intervals $\Delta_{s,i}$ contain no points of discontinuity of $\hat{u}$ and no points r_i. This can always be achieved as there is a finite number of both control points of discontinuity of $\hat{u}$ and points r_i in the package.

Let us renumber all $v_{s,i}$ and change their notation to v_i, $i \in \{1,\dots,N\}$. For brevity, we denote $\hat{x}_0 = \hat{x}(t_0)$ and $u(t,\bar{\alpha},\bar{\gamma}) = u(t,\bar{\alpha},\bar{\tau},\bar{\nu},\bar{\gamma},\bar{r},\bar{z})$. Now the control $u(t,\bar{\alpha},\bar{\gamma})$ for every $t \in [t_0,t_1]_{\mathbb{T}}$ is defined correctly by (3.4) and $u(t,\bar{\alpha},\bar{\gamma}) \in U$ for every $t \in [t_0,t_1]_{\mathbb{T}}$, because by the convexity of U, for each $i \in \{1,\dots,\nu\}$, we have

$$\hat{u}(r_i) + \sum_{p=1}^{l_i} \gamma_{p,i}\left(z_p - \hat{u}(r_i)\right) = \hat{u}(r_i)\left(1 - \sum_{p=1}^{l_i}\gamma_{p,i}\right) + \sum_{p=1}^{l_i}\gamma_{p,i} z_p \in U$$

for sufficiently small nonnegative $\gamma_{p,i}$, as a convex combination of elements from the convex set U. We renumber all $\gamma_{p,i}$ and change their notation to γ_k, $k \in \{1,\dots,\nu\}$. Obviously, $u(t,\bar{\alpha},\bar{\gamma})$ as a function of α_i and γ_k ($i \in \{1,\dots,N\}$, $k \in \{1,\dots,\nu\}$) is defined for all sufficiently small nonnegative α_k and γ_k.

Denote by $x(t,\bar{\alpha},\bar{\gamma},x_0)$ ($x(t_0,\bar{\alpha},\bar{\gamma},x_0) = x_0 \in D$) the family of phase trajectories, defined as the solution of the Cauchy problem

$$x^{\Delta}(t) = f(t,x(t),u(t,\bar{\alpha},\bar{\gamma})), \quad x(t_0) = x_0. \tag{3.5}$$

Definition 3.1: The system of linear equations of the form

$$y^{\Delta}(t) = \frac{\partial f(t,\hat{x}(t),\hat{u}(t))}{\partial x} y(t) \tag{3.6}$$

is called a system of variational equations corresponding to the optimal pair $(\hat{x},\hat{u})$.

Let us denote by $\Omega(t,s)$ a fundamental matrix of system (3.6). The following lemma plays an important rôle in the sequel.

Lemma 3.2 (About the Package of the needle-like Variations): *Let the conditions of the problem* (2.3)–(2.5) *with respect to the function f be satisfied. Suppose the conditions of density* (2.1) *are satisfied for points $s \in [t_0,t_1)_{\mathbb{T}} \cap \mathrm{RD}$. Then*

(1) *for sufficiently small $\varepsilon_0 > 0$, the solution of the Cauchy problem* (3.5) *is well defined for $t \in [t_0,t_1]_{\mathbb{T}}$ if $|x_0 - \hat{x}_0| < \varepsilon_0$, $0 \le \alpha_i < \varepsilon_0$, $0 \le \gamma_k < \varepsilon_0$, $i \in \{1,\dots,N\}$, $k \in \{1,\dots,\nu\}$;*

(2) *if $x_0 \to \hat{x}_0, \alpha_i \to 0+, \gamma_k \to 0+$ ($i \in \{1,\dots,N\}, k \in \{1,\dots,\nu\}$), then $x(t,\bar{\alpha},\bar{\gamma},x_0) \to \hat{x}(t)$ uniformly on $[t_0,t_1]_{\mathbb{T}}$;*

(3) *the mapping* $(x_0, \bar{\alpha}, \bar{\gamma}) \to x(t_1, \bar{\alpha}, \bar{\gamma}, x_0)$ *depends on* x_0 *in some neighbourhood of* $\hat{x}_0$, *and on* α_i *and* γ_k $(i \in \{1, \dots, N\}, k \in \{1, \dots, \nu\})$ *in some right neighborhood of zero, and is differentiable at the point* $(\hat{x}_0, \bar{0}, \bar{0})$, *that is, we have*

$$\begin{aligned} x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h) - \hat{x}(t_1) &= \frac{\partial x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h)}{\partial \bar{\alpha}}\Big|_{\bar{\alpha}=0, \bar{\gamma}=0, h=0} \bar{\alpha} \\ &+ \frac{\partial x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h)}{\partial \bar{\gamma}}\Big|_{\bar{\gamma}=0, \bar{\alpha}=0, h=0} \bar{\gamma} \\ &+ \frac{\partial x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h)}{\partial x_0}\Big|_{\bar{\alpha}=0, \bar{\gamma}=0, h=0} h + \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h) \end{aligned} \tag{3.7}$$

when $\bar{\alpha}, \bar{\gamma}, h \to 0$. *Here, the partial derivatives* $\frac{\partial x}{\partial \alpha_i}$ *and* $\frac{\partial x}{\partial \gamma_k}$ *are elements of the Jacobi matrices* $\frac{\partial x}{\partial \bar{\alpha}}$ *and* $\frac{\partial x}{\partial \bar{\gamma}}$ *and are understood as right-hand derivatives. Then*

$$\frac{\partial x(t_1, \bar{0}, \bar{0}, x_0)}{\partial x_0}\Big|_{x_0=\hat{x}_0} = \Omega(t_1, t_0), \tag{3.8}$$

$$\frac{\partial x(t_1, \alpha_i, \bar{0}, \hat{x}_0)}{\partial \alpha_i}\Big|_{\alpha_i=0} = \Omega(t_1, \tau_i)\left[f(\tau_i, \hat{x}(\tau_i), v_i) - f(\tau_i, \hat{x}(\tau_i), \hat{u}(\tau_i))\right], \tag{3.9}$$

$$\frac{\partial x(t_1, \bar{0}, \gamma_k, \hat{x}_0)}{\partial \gamma_k}\Big|_{\gamma_k=0} = \Omega(t_1, \sigma(r_k))\mu(r_k)\frac{\partial f(r_k, \hat{x}(r_k), \hat{u}(r_k))}{\partial u}\left(z_k - \hat{u}(r_k)\right). \tag{3.10}$$

Proof: Let us define the solution of the Cauchy problem (3.5) for sufficiently small $\bar{\alpha}, \bar{\gamma}$ and h by $x(t, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h)$. Since the admissible control $u(t, \bar{\alpha}, \bar{\gamma})$ is continuous from the right on $[t_0, t_1]_{\mathbb{T}}$, by virtue of its definition, the estimate

$$\sup_{t \in [t_0, t_1]_{\mathbb{T}}} |u(t, \bar{\alpha}, \bar{\gamma})| \le \sup_{t \in [t_0, t_1]_{\mathbb{T}}} \left|\hat{u}(t)\right| + \sum_{i=1}^{N} |v_i| + \sum_{k=1}^{\nu} |z_k| \tag{3.11}$$

is valid. From (3.4), we also obtain the existence of a function $\varphi(\bar{\alpha}, \bar{\gamma})$ such that $\varphi(\bar{\alpha}, \bar{\gamma}) \to 0$ when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$ and that the following estimate holds:

$$\int_{[t_0, t_1)_{\mathbb{T}}} \left|u(t, \bar{\alpha}, \bar{\gamma}) - \hat{u}(t)\right| \Delta t \le \varphi(\bar{\alpha}, \bar{\gamma}). \tag{3.12}$$

Now statements (1) and (2) easily follow from [29, Lemma 3] and (3.12). We now prove statement (3). Consider the increment

$$\begin{aligned} x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h) - \hat{x}(t_1) &= x(t_1, \bar{\alpha}, \bar{\gamma}, \hat{x}_0 + h) - x(t_1, 0, 0, \hat{x}_0) \\ &= x(t_1, \alpha_1, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \\ &\quad - x(t_1, 0, \alpha_2, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \\ &\quad + x(t_1, 0, \alpha_2, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \\ &\quad - x(t_1, 0, 0, \alpha_3, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \\ &\quad + \dots \\ &\quad + x(t_1, 0, 0, \dots, \alpha_k, \alpha_{k+1}, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \\ &\quad - x(t_1, 0, 0, \dots, \alpha_{k+1}, \dots, \alpha_N, \gamma_1, \dots, \gamma_\nu, \hat{x}_0 + h) \end{aligned}$$

$$
\begin{aligned}
&+\dots\\
&+x(t_1,\bar{0},\gamma_1,\dots,\gamma_\nu,\hat{x}_0+h)-x(t_1,\bar{0},0,\gamma_2,\dots,\gamma_\nu,\hat{x}_0+h)\\
&+x(t_1,\bar{0},0,\gamma_2,\dots,\gamma_\nu,\hat{x}_0+h)-x(t_1,\bar{0},0,0,\gamma_3,\dots,\gamma_\nu,\hat{x}_0+h)\\
&+\dots\\
&+x(t_1,\bar{0},0,\dots,\gamma_\nu,\hat{x}_0+h)-x(t_1,\bar{0},\bar{0},\hat{x}_0+h)\\
&+x(t_1,\bar{0},\bar{0},\hat{x}_0+h)-x(t_1,\bar{0},\bar{0},\hat{x}_0).
\end{aligned}
\tag{3.13}
$$

Thus, due to formula (3.13), consideration of the increments with respect to all of the arguments is reduced to consideration of the sum from the increments with respect to one argument.

The further proof is divided into three steps, for reader's convenience.

Step 1 At this step, we will prove differentiability of the solution $x(t_1,\bar{\alpha},\bar{\gamma},x_0)$ with respect to α_k, i.e., the needles of the first type. In (3.13), we consider the term

$$
\begin{aligned}
&x(t_1,0,\dots,0,\alpha_k,\alpha_{k+1},\dots,\alpha_N,\bar{\gamma},\hat{x}_0+h)\\
&\quad-x(t_1,0,\dots,0,\alpha_{k+1},\dots,\alpha_N,\bar{\gamma},\hat{x}_0+h).
\end{aligned}
\tag{3.14}
$$

In what follows, for the sake of brevity, we denote

$$
x_k(t)=x(t,0,\dots,0,\alpha_k,\alpha_{k+1},\dots,\alpha_N,\bar{\gamma},\hat{x}_0+h)
$$

and

$$
u_k(t)=u(t,0,\dots,\alpha_k,\alpha_{k+1},\dots,\alpha_N,\bar{\gamma}).
$$

There are two cases.

Case 1 α_k and α_{k+1} correspond to different values $\tau_k\neq\tau_{k+1}$ in (3.4). Then, according to (3.4), we have

$$
\begin{aligned}
x_k(t_1)-x_{k+1}(t_1)&=\int_{[\tau_k,\tau_k+\alpha_k)_{\mathbb{T}}}\left[f(s,x_k(s),v_k)-f(s,x_{k+1}(s),\hat{u}(s))\right]\Delta s\\
&\quad+\int_{[\tau_k+\alpha_k,t_1)_{\mathbb{T}}}\left[f(s,x_k(s),u_{k+1}(s))-f(s,x_{k+1}(s),u_{k+1}(s))\right]\Delta s\\
&=I_1+I_2.
\end{aligned}
\tag{3.15}
$$

Consider each integral in (3.15) separately. First,

$$
\begin{aligned}
I_1=&\int_{[\tau_k,\tau_k+\alpha_k)_{\mathbb{T}}}\left[f(s,x_k(s),v_k)-f(s,\hat{x}(s),v_k)\right]\Delta s\\
&+\int_{[\tau_k,\tau_k+\alpha_k)_{\mathbb{T}}}\left[f(s,\hat{x}(s),v_k)-f(s,\hat{x}(s),\hat{u}(s))\right]\Delta s\\
&+\int_{[\tau_k,\tau_k+\alpha_k)_{\mathbb{T}}}\left[f(s,\hat{x}(s),\hat{u}(s))-f(s,x_{k+1}(s),\hat{u}(s))\right]\Delta s\\
=&I_{11}+I_{12}+I_{13}.
\end{aligned}
\tag{3.16}
$$

Statement (2) implies the existence of a function $\varphi_k(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h) \to 0$ when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$ such that

$$|I_{11}| + |I_{13}| \leq \varphi_k(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)\, |[\tau_k, \tau_k + \alpha_k)_{\mathbb{T}}|\,, \tag{3.17}$$

where $|\cdot|$ indicates the length of the interval under consideration. Let $\tau_k + \alpha_k'$ be the nearest point of the time scale at the left of the point $\tau_k + \alpha_k$. Then $|[\tau_k, \tau_k + \alpha_k)_{\mathbb{T}}| = \alpha_k'$. But $\alpha_k' \leq \alpha_k \leq \alpha_k' + \mu(\tau_k + \alpha_k')$. Hence

$$1 \leq \frac{\alpha_k}{\alpha_k'} \leq 1 + \frac{\mu(\tau_k + \alpha_k')}{\alpha_k'}.$$

Therefore, considering condition (2.1), we have

$$\frac{\alpha_k'}{\alpha_k} \to 1 \quad \text{when} \quad \alpha_k \to 0. \tag{3.18}$$

Then, from (3.17), it follows that

$$|I_{11}| + |I_{13}| = \mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h) \quad \text{when} \quad \bar{\alpha} \to 0,\ \bar{\gamma} \to 0,\ h \to 0. \tag{3.19}$$

Let us consider the term I_{12} in (3.16). We have

$$\begin{aligned} I_{12} = &\int_{[\tau_k, \tau_k + \alpha_k)_{\mathbb{T}}} \left[f(s, \hat{x}(s), v_k) - f(\tau_k, \hat{x}(\tau_k), v_k)\right] \Delta s \\ &+ \int_{[\tau_k, \tau_k + \alpha_k)_{\mathbb{T}}} \left[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))\right] \Delta s \\ &+ \int_{[\tau_k, \tau_k + \alpha_k)_{\mathbb{T}}} \left[f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k)) - f(s, \hat{x}(s), \hat{u}(s))\right] \Delta s \\ = &\, I_{21} + I_{22} + I_{23}. \end{aligned} \tag{3.20}$$

We note that the points $(s, \hat{x}(s), v_k)$ and $(s, \hat{x}(s), \hat{u}(s))$ belong to a common compact set K for all $s \in [t_0, t_1]_{\mathbb{T}}$ and for all v_k, $k \in \{1, \ldots, N\}$. Then, from the continuity of $\hat{x}(s)$ and from the uniform continuity of $f(t, x, v)$ on the indicated compact set and (3.18), it follows that $|I_{21}| = \mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)$. Taking into account that the point τ_k is a point of piecewise continuity of the control $\hat{u}$, we similarly have that $|I_{23}| = \mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)$. From (3.18), we obtain that

$$\begin{aligned} I_{22} = &\left[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))\right] \alpha_k \\ &+ \left[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))\right] (\alpha_k' - \alpha_k). \end{aligned}$$

By virtue of (3.18), $\alpha_k' - \alpha_k = \mathrm{o}\,(\alpha_k)$ as $\alpha_k \to 0$. Therefore

$$I_{22} = \left[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))\right] \alpha_k + \mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h) \tag{3.21}$$

when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. From (3.16), (3.19), and (3.21), it follows that the first term in (3.15) takes the form

$$[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))]\alpha_k + \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h) \tag{3.22}$$

when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. Consider now the second term in (3.15). We obtain

$$I_2 = \int_{[\tau_k+\alpha_k, t_1)_{\mathbb{T}}} \frac{\partial f}{\partial x}\left(s, \theta(s, \bar{\alpha}, \bar{\gamma}, h), u_{k+1}(s)\right)\left(x_k(s) - x_{k+1}(s)\right)\Delta s. \tag{3.23}$$

The last equality is obtained using Lagrange's formula. Here,

$$\theta(t, \bar{\alpha}, \bar{\gamma}, h) = x_{k+1}(t) + \theta(x_k(t) - x_{k+1}(t)) \in \mathbb{R}^d \quad \text{for} \quad \theta \in [0, 1],$$

where θ depends on $t, \bar{\alpha}, \bar{\gamma}, h$. However, as follows from [29, Proposition 2], the expression $\frac{x_k(t)-x_{k+1}(t)}{\alpha_k}$ converges uniformly to $\left.\frac{\partial x_k(t)}{\partial \alpha_k}\right|_{\alpha_k=0}$ on $[\tau_k, t_1]_{\mathbb{T}}$ as $\alpha_k \to 0$, where $\left.\frac{\partial x_k(t)}{\partial \alpha_k}\right|_{\alpha_k=0}$ is a solution of the linear system

$$y^{\Delta}(t) = \frac{\partial f}{\partial x}(t, x_{k+1}(t), u_{k+1}(t))y(t) \tag{3.24}$$

on $[\tau_k, t_1]_{\mathbb{T}}$ with the initial condition

$$\left.\frac{\partial x_k(\tau_k)}{\partial \alpha_k}\right|_{\alpha_k=0} = f(\tau_k, x_{k+1}(\tau_k), v_k) - f(\tau_k, x_{k+1}(\tau_k), u_{k+1}(\tau_k)). \tag{3.25}$$

Let us show that the solution of the Cauchy problem (3.24), (3.25) converges uniformly on $[\tau_k, t_1)_{\mathbb{T}}$ to the solution of the adjoint system (3.6) with the initial condition

$$y(\tau_k) = f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k)) \tag{3.26}$$

when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. Denote the solution of the Cauchy problem (3.24), (3.25) by y_k, and denote the solution of the Cauchy problem (3.6), (3.26) by y. We note that as long as $u_{k+1}(t) = \hat{u}(t)$ when $t \in [t_0, \tau_{k+1})_{\mathbb{T}}$, then $x_{k+1}(t) = \hat{x}(t)$ when $t \in [t_0, \tau_{k+1})_{\mathbb{T}}$. Therefore, the initial conditions (3.25) and (3.26) coincide. We have

$$\begin{aligned}
|y_k(t) - y(t)| &\le \int_{[\tau_{k+1}, t_1)_{\mathbb{T}}} \left|\frac{\partial f}{\partial x}\left(s, x_{k+1}(s), u_{k+1}(s)\right) y_k(s) - \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s))y(s)\right| \Delta s \\
&\le \int_{[\tau_{k+1}, t_1)_{\mathbb{T}}} \left|\frac{\partial f}{\partial x}(s, x_{k+1}(s), u_{k+1}(s))\right| \left|y_k(s) - y(s)\right| \Delta s \\
&\quad + \int_{[\tau_{k+1}, t_1)_{\mathbb{T}}} \left|\frac{\partial f}{\partial x}(s, x_{k+1}(s), u_{k+1}(s)) - \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s))\right| \left|y(s)\right| \Delta s \\
&= I_1 + I_2.
\end{aligned} \tag{3.27}$$

Let us show that I_2 tends to zero when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. Indeed, as $y(t)$ is bounded on $[t_0, t_1]_{\mathbb{T}}$, it suffices to show that

$$\int_{[\tau_{k+1},t_1)_{\mathbb{T}}} \left| \frac{\partial f}{\partial x}(s, x_{k+1}(s), u_{k+1}(s)) - \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s)) \right| \Delta s \to 0 \tag{3.28}$$

when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. From (3.11), the continuity of $\frac{\partial f}{\partial x}$ in its arguments, and the statement (2), it follows that the point $(s, x_{k+1}(s), u_{k+1}(s))$ belongs to the above mentioned compact set K for all $s \in [t_0, t_1]_{\mathbb{T}}$, for sufficiently small $\bar{\alpha}, \bar{\gamma}, h$. Therefore, $\frac{\partial f}{\partial x}(t, x, v)$ is uniformly continuous on its arguments on this compact set. Inequality (3.12) implies a reasonable convergence of $u_{k+1}(t)$ to $\hat{u}(t)$ when $\bar{\alpha} \to 0$ and $\bar{\gamma} \to 0$. Hence, taking into account the uniform continuity of $\frac{\partial f}{\partial x}$, we conclude that the integrand in (3.28) tends to zero when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. Since, for sufficiently small $\bar{\alpha}, \bar{\gamma}, h$, it is uniformly bounded in these arguments, by Lebesgue's bounded convergence theorem [21, p. 330], (3.28) holds. Therefore, from (3.27) and an analogue of Gronwall's lemma [29, Lemma 2], we have

$$\sup_{t \in [t_0,t_1]_{\mathbb{T}}} \left| y_k(t) - y(t) \right| \to 0 \quad \text{when} \quad \bar{\alpha} \to 0,\ \bar{\gamma} \to 0,\ h \to 0. \tag{3.29}$$

Further, we have

$$\begin{aligned}
&\alpha_k \left| I_2 - \int_{[\tau_k,t_1)_{\mathbb{T}}} \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s)) y(s) \Delta s \right| \\
&\le \alpha_k \int_{[\tau_k,\tau_k+\alpha_k)_{\mathbb{T}}} \left| \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s)) y(s) \right| \Delta s \\
&\quad + \alpha_k \int_{[\tau_k+\alpha_k,t_1)_{\mathbb{T}}} \left| \frac{\partial f}{\partial x}(s, \theta(s, \bar{\alpha}, \bar{\gamma}, h), u_{k+1}(s)) \right| \left| \frac{x_k(s) - x_{k+1}(s)}{\alpha_k} - y(s) \right| \Delta s \\
&\quad + \alpha_k \int_{[\tau_k+\alpha_k,t_1)_{\mathbb{T}}} \left| \frac{\partial f}{\partial x}(s, \theta(s, \bar{\alpha}, \bar{\gamma}, h), u_{k+1}(s)) - \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s)) \right| \left| y(s) \right| \Delta s.
\end{aligned} \tag{3.30}$$

The first term in (3.30) is obviously equal to $\mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)$ when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$. Taking into account [29, Proposition 2], (3.29), and the boundedness of $\frac{\partial f}{\partial x}(s, \theta(s, \bar{\alpha}, \bar{\gamma}, h), u_{k+1}(s))$, we conclude that the second term in (3.30) is $\mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)$. Similarly to (3.28), it is now easy to show that the third term in (3.30) is $\mathrm{o}\,(\alpha_k, \ldots, \alpha_N, \bar{\gamma}, h)$ as well. Returning now to the formula (3.15) and using (3.22) and (3.30) for the increment $x_k(t_1) - x_{k+1}(t_1)$, we have

$$\begin{aligned}
x_k(t_1) - x_{k+1}(t_1) &= \left[f(\tau_k, \hat{x}(\tau_k), v_k) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k)) \right] \alpha_k \\
&\quad + \int_{[\tau_k,t_1)_{\mathbb{T}}} \frac{\partial f}{\partial x}(s, \hat{x}(s), \hat{u}(s)) y(s) \Delta s \cdot \alpha_k + \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h) \\
&= \left. \frac{\partial x(t_1, \alpha_k, \bar{0}, \hat{x}_0)}{\partial \alpha_k} \right|_{\alpha_k = 0} \cdot \alpha_k + \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h)
\end{aligned} \tag{3.31}$$

when $\bar{\alpha} \to 0$, $\bar{\gamma} \to 0$, $h \to 0$.

Case 2 α_k and α_{k+1} correspond to the same τ_k. Assume, for simplicity, that the considered package of needle-like variations $(\bar{\alpha}, \bar{\tau}, \bar{v}, \bar{\gamma}, \bar{r}, \bar{z})$ contains three kinds of needle variations

(τ_k, ν_k), (τ_k, ν_{k+1}), and (τ_k, ν_{k+2}), corresponding to the same τ_k. For a different number of needle variations, the argument is similar. Let us consider again the term in (3.13)

$$\begin{aligned} x_k(t_1) - x_{k+1}(t_1) = x(t_1, 0, 0, \dots, 0, \alpha_k, \alpha_{k+1}, \alpha_{k+2}, \dots, \alpha_N, \bar{\gamma}, \hat{x}_0 + h) \\ - x(t_1, 0, \dots, 0, \alpha_{k+1}, \alpha_{k+2}, \dots, \alpha_N, \bar{\gamma}, \hat{x}_0 + h). \end{aligned} \tag{3.32}$$

Now the action of the needle-like variation (τ_k, ν_{k+1}) in the control $u_{k+1}(t) = u(t, 0, \dots, 0, \alpha_{k+1}, \dots, \alpha_N, \bar{\gamma})$ will be shifted to the left by α_k, and the action of the needle-like variation (τ_k, ν_{k+2}) will be shifted to the left by $2\alpha_k$ compared to their action in the control $u_k(t) = u(t, 0, \dots, 0, \alpha_k, \alpha_{k+1}, \dots, \alpha_N, \bar{\gamma})$. The operation of the rest of the needle variations will not be shifted. Based on the control $u_{k+1}(t)$, we build a new admissible control $u_{sh}(t)$, where the action of the needle-like variation (τ_k, ν_{k+1}) is shifted to the right by α_k, and the action of the needle-like variation (τ_k, ν_{k+2}) is shifted to the right by $2\alpha_k$:

$$u_{sh}(t) = \begin{cases} \nu_{k+1} & \text{if } t \in [\tau_k + \alpha_k + \alpha_{k+1} + \alpha_{k+2}, \tau_k + \alpha_k + \alpha_{k+2} + 2\alpha_{k+1})_{\mathbb{T}}, \\ \nu_{k+2} & \text{if } t \in [\tau_k + 2(\alpha_k + \alpha_{k+1} + \alpha_{k+2}), \\ & \quad \tau_k + 2(\alpha_k + \alpha_{k+1} + \alpha_{k+2}) + \alpha_{k+2})_{\mathbb{T}}, \\ \hat{u}(t) & \text{if } t \in [\tau_k, \tau_k + \alpha_k + \alpha_{k+1} + \alpha_{k+2})_{\mathbb{T}} \\ & \quad \cup [\tau_k + \alpha_k + \alpha_{k+2} + 2\alpha_{k+1}, \tau_k + 2(\alpha_k + \alpha_{k+1} + \alpha_{k+2}))_{\mathbb{T}}. \end{cases}$$

Let us put, more precisely, $\alpha = \alpha_k + \alpha_{k+1} + \alpha_{k+2}$ and consider a control

$$u_{sh}(t) = \begin{cases} \hat{u}(t) & \text{if } t \in [\tau_k, \tau_k + \alpha)_{\mathbb{T}} \cup [\tau_k + \alpha + \alpha_{k+1}, \tau_k + 2\alpha)_{\mathbb{T}}, \\ \nu_{k+1} & \text{if } t \in [\tau_k + \alpha, \tau_k + \alpha + \alpha_{k+1})_{\mathbb{T}}, \\ \nu_{k+2} & \text{if } t \in [\tau_k + 2\alpha, \tau_k + 2\alpha + \alpha_{k+2})_{\mathbb{T}}, \\ u_{k+1}(t) & \text{elsewhere.} \end{cases} \tag{3.33}$$

Now in the controls $u_k(t)$ and $u_{sh}(t)$, the needle-like variations (τ_k, ν_{k+1}) and (τ_k, ν_{k+2}) operate at regular intervals. Let $x_{sh}(t)$ be a solution of the Cauchy problem (3.5) with $x_0 = \hat{x}_0 + h$, corresponding to the control (3.32). We have

$$x_k(t_1) - x_{k+1}(t_1) = x_k(t_1) - x_{sh}(t_1) + x_{sh}(t_1) - x_{k+1}(t_1). \tag{3.34}$$

All reasoning from Case 1 with different τ_k and τ_{k+1} is applicable to the first term in (3.34). Therefore, we have

$$\begin{aligned} x_k(t_1) - x_{sh}(t_1) = \left.\frac{\partial x(t_1, \alpha_k, \bar{0}, \hat{x}_0)}{\partial \alpha_k}\right|_{\alpha_k = 0} \alpha_k + o(\bar{\alpha}, \bar{\gamma}, h) \\ \text{when} \quad \bar{\alpha} \to 0, \ \bar{\gamma} \to 0, \ h \to 0. \end{aligned} \tag{3.35}$$

Let us show that

$$x_{sh}(t_1) - x_{k+1}(t_1) = o(\bar{\alpha}, \bar{\gamma}, h) \quad \text{when} \quad \bar{\alpha} \to 0, \ \bar{\gamma} \to 0, \ h \to 0. \tag{3.36}$$

According to (3.33), we have

$$
\begin{aligned}
&x_{sh}(t_1) - x_{k+1}(t_1) \\
&\quad = \int_{[\tau_k+\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha)_{\mathbb{T}}} \left[f(s, x_{sh}(s), \hat{u}(s)) - f(s, x_{k+1}(s), v_{k+1})\right] \Delta s \\
&\qquad + \int_{[\tau_k+\alpha,\tau_k+2\alpha_{k+1}+\alpha_{k+2})_{\mathbb{T}}} \left[f(s, x_{sh}(s), v_{k+1}) - f(s, x_{k+1}(s), v_{k+1})\right] \Delta s \\
&\qquad + \int_{[\tau_k+2\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha_k+\alpha_{k+2}+2\alpha_{k+1})_{\mathbb{T}}} \left[f(s, x_{sh}(s), v_{k+1})\right. \\
&\qquad \left. - f(s, x_{k+1}(s), \hat{u}(s))\right] \Delta s \\
&\qquad + \int_{[\tau_k+\alpha+\alpha_{k+1},\tau_k+2(\alpha_{k+1}+\alpha_{k+2}))_{\mathbb{T}}} \left[f(s, x_{sh}(s), \hat{u}(s)) - f(s, x_{k+1}(s), \hat{u}(s))\right] \Delta s \\
&\qquad + \int_{[\tau_k+2(\alpha_{k+1}+\alpha_{k+2}),\tau_k+2\alpha)_{\mathbb{T}}} \left[f(s, x_{sh}(s), \hat{u}(s)) - f(s, x_{k+1}(s), v_{k+2})\right] \Delta s \\
&\qquad + \int_{[\tau_k+2\alpha,\tau_k+2(\alpha_{k+1}+\alpha_{k+2})+\alpha_{k+2})_{\mathbb{T}}} \left[f(s, x_{sh}(s), v_{k+2})\right. \\
&\qquad \left. - f(s, x_{k+1}(s), v_{k+2})\right] \Delta s \\
&\qquad + \int_{[\tau_k+2(\alpha_{k+1}+\alpha_{k+2})+\alpha_{k+2},\tau_k+2\alpha+\alpha_{k+2})_{\mathbb{T}}} \left[f(s, x_{sh}(s), v_{k+2})\right. \\
&\qquad \left. - f(s, x_{k+1}(s), \hat{u}(s))\right] \Delta s \\
&\qquad + \int_{[\tau_k+2\alpha+\alpha_{k+2},t_1)_{\mathbb{T}}} \left[f(s, x_{sh}(s), \hat{u}(s)) - f(s, x_{k+1}(s), \hat{u}(s))\right] \Delta s \\
&\quad = \sum_{i=1}^{8} I_i.
\end{aligned}
\tag{3.37}
$$

Let us consider the terms in (3.37) separately. In assessing the first term, for the sake of brevity, we substitute the symbol $\int_{[\tau_k+\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha)} \cdot \Delta s$ by $\int \cdot \Delta s$. Then we have

$$
\begin{aligned}
I_1 = &\int \left[f(s, x_{sh}(s), \hat{u}(s)) - f(s, \hat{x}(s), \hat{u}(s))\right] \Delta s \\
&+ \int \left[f(s, \hat{x}(s), \hat{u}(s)) - f(s, \hat{x}(s), v_{k+1})\right] \Delta s \\
&+ \int \left[f(s, \hat{x}(s), v_{k+1}) - f(s, x_{k+1}(s), v_{k+1})\right] \Delta s \\
=&I_{11} + I_{12} + I_{13}.
\end{aligned}
\tag{3.38}
$$

By virtue of statement (2), we have

$$
I_{11} + I_{13} = \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h) \quad \text{when} \quad \bar{\alpha} \to 0,\ \bar{\gamma} \to 0,\ h \to 0. \tag{3.39}
$$

Next,

$$\begin{aligned} I_{12} = &\int \left[f(s,\hat{x}(s),\hat{u}(s)) - f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k))\right]\Delta s \\ &+ \int \left[f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k)) - f(\tau_k,\hat{x}(\tau_k),v_{k+1})\right]\Delta s \\ &+ \int \left[f(\tau_k,\hat{x}(\tau_k),v_{k+1}) - f(s,\hat{x}(s),v_{k+1})\right]\Delta s. \end{aligned} \tag{3.40}$$

By the piecewise continuity of $\hat{u}(s)$ at the point τ_k, the first and third terms in (3.40) are equal to $\mathrm{o}\,(\bar{\alpha},\bar{\gamma},h)$. Let us consider the second term in (3.40). It is equal to

$$\left[f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k)) - f(\tau_k,\hat{x}(\tau_k),v_{k+1})\right]\cdot\left|[\tau_k+\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha)_{\mathbb{T}}\right|, \tag{3.41}$$

where, as before, the symbol $|\cdot|$ indicates the length of the interval. The nearest right point of the time scale to $\tau_k+\alpha_{k+1}+\alpha_{k+2}$ is located at a distance $\mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2})$ from it by virtue of condition (2.1) on the time scale. If $\alpha_k = \mathrm{o}(\alpha_{k+1}+\alpha_{k+2})$, then the expression (3.41) is $\mathrm{o}\,(\bar{\alpha},\bar{\gamma},h)$, and I_1 in (3.38) is equal to $\mathrm{o}\,(\bar{\alpha},\bar{\gamma},h)$. Let $\alpha_k \neq \mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2})$. Then there are several points of the time scale in the interval $[\tau_k+\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha)_{\mathbb{T}}$. Indeed, the nearest right point of the time scale to $\tau_k+\alpha_{k+1}+\alpha_{k+2}$ is $\tau_k+\alpha_{k+1}+\alpha_{k+2}+\mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2})$, and the next point of the time scale after $\tau_k+\alpha_{k+1}+\alpha_{k+2}+\mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2})$ is situated at a distance not greater than $\mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2}+\mathrm{o}\,(\alpha_{k+1}+\mathrm{o}\,(\alpha_{k+2}))) = \mathrm{o}\,(\alpha_{k+1}+\alpha_{k+2})$. Let the point $s=\tau_k+\beta$ be the largest point of the time scale before the point $\tau_k+\alpha$, i.e., $\tau_k+\beta \le \tau_k+\alpha$. If $\tau_k+\beta \neq \tau_k+\alpha$, then the next point of the time scale is $\tau_k+\beta+\mathrm{o}\,(\beta)$. But $\mathrm{o}\,(\beta) = \mathrm{o}\,(\alpha_k+\alpha_{k+1}+\alpha_{k+2})$ when $\bar{\alpha}\to 0$. Indeed

$$\frac{\mathrm{o}\,(\beta)}{\alpha_k+\alpha_{k+1}+\alpha_{k+2}} = \frac{\mathrm{o}\,(\beta)}{\beta}\frac{\beta}{\alpha_k+\alpha_{k+1}+\alpha_{k+2}} \to 0 \quad \text{when} \quad \bar{\alpha}\to 0.$$

Note that $\beta\to 0$ when $\bar{\alpha}\to 0$. Hence $s=\tau_k+\alpha-\mathrm{o}\,(\alpha_k+\alpha_{k+1}+\alpha_{k+2})$. Therefore, the length of the interval is

$$\begin{aligned} \left|[\tau_k+\alpha_{k+1}+\alpha_{k+2},\tau_k+\alpha)_{\mathbb{T}}\right| &= \tau_k+\alpha-\mathrm{o}\,(\bar{\alpha})-\tau_k-\alpha_{k+1}-\alpha_{k+2}-\mathrm{o}\,(\bar{\alpha}) \\ &= \alpha_k+\mathrm{o}\,(\bar{\alpha}) \quad \text{as} \quad \bar{\alpha}\to 0. \end{aligned}$$

Then, from the uniform boundedness of $f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k)) - f(\tau_k,\hat{x}(\tau_k),v_{k+1})$, we have that in (3.40),

$$I_{12} = \left[f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k)) - f(\tau_k,\hat{x}(\tau_k),v_{k+1})\right]\alpha_k + \mathrm{o}\,(\bar{\alpha}) \quad \text{when} \quad \bar{\alpha}\to 0.$$

Consequently, the term I_1 in (3.36) is equal to

$$I_1 = \left[f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k)) - f(\tau_k,\hat{x}(\tau_k),v_{k+1})\right]\alpha_k + \mathrm{o}\,(\bar{\alpha}) \quad \text{when} \quad \bar{\alpha}\to 0. \tag{3.42}$$

It is easy to show that in (3.37), $I_2 = \mathrm{o}\,(\bar{\alpha})$. Let us consider the term I_3. Similar to the estimate of I_1, we can show that

$$I_3 = \left[f(\tau_k,\hat{x}(\tau_k),v_{k+1}) - f(\tau_k,\hat{x}(\tau_k),\hat{u}(\tau_k))\right]\alpha_k + \mathrm{o}\,(\bar{\alpha}) \quad \text{when} \quad \bar{\alpha}\to 0. \tag{3.43}$$

It is obvious that $I_4 = \mathrm{o}(\bar{\alpha})$ and

$$I_5 = \left[f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k)) - f(\tau_k, \hat{x}(\tau_k), v_{k+2})\right] 2\alpha_k + \mathrm{o}(\bar{\alpha}), \tag{3.44}$$

$I_6 = \mathrm{o}(\bar{\alpha})$, and

$$I_7 = \left[f(\tau_k, \hat{x}(\tau_k), v_{k+2}) - f(\tau_k, \hat{x}(\tau_k), \hat{u}(\tau_k))\right] 2\alpha_k + \mathrm{o}(\bar{\alpha}). \tag{3.45}$$

Then, from (3.37), (3.42)–(3.45), we have that in (3.37)

$$\sum_{i=1}^{7} I_i = \mathrm{o}(\bar{\alpha}) \quad \text{as} \quad \alpha \to 0. \tag{3.46}$$

From (3.37), we have

$$\left|x_{sh}(t_1) - x_{k+1}(t_1)\right| \leq \mathrm{o}(\bar{\alpha}) + \int_{[\tau_k+2\alpha+\alpha_{k+2}, t_1)_{\mathbb{T}}} L \left|x_{sh}(s) - x_{k+1}(s)\right| \Delta s, \tag{3.47}$$

where L is the Lipschitz constant for f in the compact set K. Using Gronwall's lemma [29, Lemma 2], we get (3.36). Hence, using (3.34) and (3.35), we get

$$x_k(t_1) - x_{k+1}(t_1) = \frac{\partial x(t_1, \alpha_k, \bar{0}, \hat{x}_0)}{\partial \alpha_k}\Big|_{\alpha_k=0} \alpha_k + \mathrm{o}(\bar{\alpha}, \bar{\gamma}, h) \quad \text{as} \quad \bar{\alpha} \to 0,\ \bar{\gamma} \to 0,\ h \to 0. \tag{3.48}$$

Now, taking into account (3.31), we conclude that in the case of a needle-like variation that corresponds to τ_k and in the case of multiple needle-like variations, we have the representation (3.48).

Step 2 Within this step, we will prove differentiability of the solutions $x(t_1, \bar{\alpha}, \bar{\gamma}, x_0)$ with respect to γ_k, i.e., the needles of the second type.

Let us now consider the second type of needle-like variations. Let us consider in (3.13) increments of the form

$$x(t_1, \bar{0}, 0, \ldots, 0, \gamma_k, \gamma_{k+1}, \ldots, \gamma_\nu, \hat{x}_0 + h) - x(t_1, \bar{0}, 0, \ldots, 0, \gamma_{k+1}, \ldots, \gamma_\nu, \hat{x}_0 + h). \tag{3.49}$$

Similarly as before, we again denote by

$$x_k(t) = x(t_1, \bar{0}, 0, \ldots, 0, \gamma_k, \gamma_{k+1}, \ldots, \gamma_\nu, \hat{x}_0 + h)$$

a solution of the Cauchy problem (3.5) with $x_k(t_0) = \hat{x}_0 + h$, satisfying an admissible control $u_k(t) = u(t, \bar{0}, \gamma_k, \ldots, \gamma_\nu)$. Let a few needle-like variations correspond to some r_k in our package. Without loss of generality, we assume for simplicity that the initial package

contains two needle-like variations (r_k, z_1) and (r_k, z_2). Then

$$\begin{aligned} x_k(t_1) - x_{k+1}(t_1) &= \int_{[r_k,t_1)_{\mathbb{T}}} [f(s, x_k(s), u_k(s)) - f(s, x_{k+1}(s), u_{k+1}(s))]\Delta s \\ &= \mu(r_k)[f(r_k, x_k(r_k), u_k(r_k)) - f(r_k, x_{k+1}(r_k), u_{k+1}(r_k))] \\ &\quad + \int_{[\sigma(r_k),t_1)_{\mathbb{T}}} [f(s, x_k(s), u_k(s)) - f(r_k, x_{k+1}(s), u_{k+1}(s))]\Delta s. \end{aligned}$$

In the last formula, we note that $x_k(r_k) = x_{k+1}(r_k) = \hat{x}(r_k)$ and $u_k(s) = u_{k+1}(s)$ when $s \in [\sigma(\tau_k), t_1)_{\mathbb{T}}$. Then

$$\begin{aligned} x_k(t_1) - x_{k+1}(t_1) &= \mu(r_k)[f(r_k, \hat{x}(r_k), u_k(r_k)) - f(r_k, \hat{x}(r_k), u_{k+1}(r_k))] \\ &\quad + \int_{[\sigma(r_k),t_1)_{\mathbb{T}}} [f(s, x_k(s), u_{k+1}(s)) - f(s, x_{k+1}(s), u_{k+1}(s))]\Delta s. \end{aligned} \tag{3.50}$$

The first term in (3.50) is, according to Lagrange's formula,

$$\mu(r_k)\left[\frac{\partial f}{\partial u}(r_k, \hat{x}(r_k), \theta(r_k, \bar{\gamma}))\gamma_k(z_1 - \hat{u}(r_k))\right], \tag{3.51}$$

where $\theta(r_k, \bar{\gamma}) = u_{k+1}(r_k) + \theta(u_k(r_k) - u_{k+1}(r_k))$ for some $\theta \in [0, 1]$. However, from the uniform continuity of $\frac{\partial f}{\partial u}$ on K, the expression $\frac{\partial f}{\partial u}(r_k, \hat{x}(r_k), \theta(r_k, \bar{\gamma}))$ tends uniformly to $\frac{\partial f}{\partial u}(r_k, \hat{x}(r_k), \hat{u}(r_k))$ for all r_k and z_k when $\bar{\gamma} \to 0$, $h \to 0$. Therefore, (3.51) takes the form

$$\mu(r_k)\frac{\partial f}{\partial u}(r_k, \hat{x}(r_k), \hat{u}(r_k))(z_1 - \hat{u}(r_k))\gamma_k + \mathrm{o}(\bar{\gamma}, h) \quad \text{when} \quad \bar{\gamma} \to 0,\ h \to 0. \tag{3.52}$$

Let us consider the second term in (3.50):

$$\int_{[\sigma(\tau_k),t_1)_{\mathbb{T}}} \frac{\partial f}{\partial x}(s, \theta(s, \bar{\gamma}, h), u_{k+1}(s))\frac{x_k(s) - x_{k+1}(s)}{\gamma_k}\Delta s \cdot \gamma_k. \tag{3.53}$$

However, as it follows from [29, Proposition 1], the expression $\frac{x_k(t)-x_{k+1}(t)}{\gamma_k}$ converges uniformly to $\left.\frac{\partial x_k(t)}{\partial \gamma_k}\right|_{\gamma_k=0}$ on $[\sigma(\tau_k), t_1)_{\mathbb{T}}$ as $\alpha_k \to 0$. Here, $\left.\frac{\partial x_k}{\partial \gamma_k}\right|_{\gamma_k=0}$ is a solution on $[\sigma(\tau_k), t_1)_{\mathbb{T}}$ of the linear system

$$y^{\Delta}(t) = \frac{\partial f}{\partial x}(t, x_{k+1}(t), u_{k+1}(t))y(t)$$

with initial condition

$$y(\sigma(r_k)) = \mu(r_k)\frac{\partial f}{\partial u}(r_k, \hat{x}(r_k), \hat{u}(r_k))(z_1 - \hat{u}(r_k)). \tag{3.54}$$

Verifying now arguments which are similar to calculations with a single needle-like variation of the first kind (formulas (3.27)–(3.30)), we obtain

$$x_k(t_1) - x_{k+1}(t_1) = \frac{\partial x(t_1, \bar{\alpha}, \gamma_k, \hat{x}_0)}{\partial \gamma_k}\Big|_{\gamma_k=0} \gamma_k + \mathrm{o}\,(\bar{\alpha}, \bar{\gamma}, h) \quad \text{when} \quad \bar{\alpha} \to 0,\ \bar{\gamma} \to 0,\ h \to 0. \tag{3.55}$$

Step 3 Let us prove differentiability of the solutions $x(t_1, \bar{\alpha}, \bar{\gamma}, x_0)$ with respect to x_0.

Let us consider the last term in (3.13). As it follows from [43], the solution $x(t, \bar{0}, \bar{0}, \hat{x}_0+h)$ is continuously differentiable for h at the point $h = 0$, and therefore

$$x(t_1, \bar{0}, \bar{0}, \hat{x}_0 + h) - \hat{x}(t_1) = \frac{\partial x(t_1, \bar{0}, \bar{0}, \hat{x}_0 + h)}{\partial h}\Big|_{h=0} h + \mathrm{o}\,(\,|h|\,) \quad \text{as} \quad h \to 0. \tag{3.56}$$

Considering now (3.31), (3.48), (3.55), and (3.56), we get (3.7) from (3.13). For left-dense points, the arguments are similar, but in the needle-like variations of the first kind, the intervals $\Delta_{s,i}$ should be chosen as

$$\Delta_{s,i} = [\tau_i - (s-1)\bar{\alpha}_i - \alpha_{s,i}, \tau_i - (s-1)\bar{\alpha}_i)_{\mathbb{T}}.$$

The proof is complete. □

Related to Lemma 3.2, we give the following two remarks.

Remark 3.3: All the statements of Lemma 3.2 obviously remain valid for packages containing among needle-like variations of the first kind both left-dense and right-dense points.

Remark 3.4: In the proof of a similar result for ordinary differential equations, they use the fact that the function $x(t_1, \bar{\alpha}, \bar{\gamma}, x_0)$ is continuously differentiable in some neighbourhood of the point $(\bar{0}, \bar{0}, \hat{x}_0)$. In our case, the function $x(t_1, \bar{\alpha}, \bar{\gamma}, x_0)$ has no partial derivatives w.r.t. α_i when $\alpha_i \neq 0$. This is the main difference of our case from the classical lemma for a package of needle-like variations and the main difficulty in its proof.

3.2. Necessary conditions for an extreme in the cone

In this subsection, for the sake of completeness, we give the necessary extreme conditions for smooth problems with equality and inequality-type constraints that are given in a cone.

Let U be a neighbourhood of a point $\hat{x} \in \mathbb{R}^n$. We denote by $\hat{x} + \mathbb{R}^n_+$ the set

$$\hat{x} + \mathbb{R}^n_+ = \{\hat{x} + \alpha :\ \alpha = (\alpha_1, \ldots, \alpha_n),\ \alpha_i \geq 0,\ i \in \{1, \ldots, n\}\}.$$

Let the functions f_i, $i \in \{0, \ldots, m\}$, be defined on the set $U \cap (\hat{x} + \mathbb{R}^n_+)$. Let us consider the problem

$$\begin{aligned} f_0(x) &\to \inf, \quad x \in U \cap (\hat{x} + \mathbb{R}^n_+), \\ f_i(x) &\leq 0, \quad i \in \{1, \ldots, m'\}, \\ f_i(x) &= 0, \quad i \in \{m' + 1, \ldots, m\}. \end{aligned} \tag{3.57}$$

We have the following theorem.

Theorem 3.5 (See [3,6]): *Suppose that in the problem (3.57), the functions f_i, $i \in \{0, \dots, m\}$ are continuous on $U \cap (\hat{x} + \mathbb{R}^n_+)$ and differentiable at $\hat{x}$ relative to the set $(\hat{x} + \mathbb{R}^n_+)$. If $\hat{x}$ is a local extreme in (3.57), then there exists a nonzero vector of Lagrange multipliers $\lambda = (\lambda_0, \lambda_1, \dots, \lambda_m)$ such that*

(a) $\lambda_i \geq 0$, $i \in \{0, \dots, m\}$;
(b) $\lambda_i f_i(\hat{x}) = 0$, $i \in \{1, \dots, m'\}$;
(c) $\sum_{i=0}^{m} \lambda_i f_i'(\hat{x} + 0) \geq 0$,

where $f_i'(\hat{x} + 0)$ is the right derivative of the function f_i at the point $\hat{x}$, $i \in \{0, \dots, m\}$.

3.3. About adjoint systems

In this subsection, we present a result of the correspondence between solutions of the linear system

$$x^{\Delta}(t) = A(t)x(t) \tag{3.58}$$

and its adjoint system

$$y^{\Delta}(t) = -A^*(t)y(\sigma(t)), \tag{3.59}$$

where the $d \times d$-matrix-valued function A is μ_Δ-measurable on $[t_0, t_1]_{\mathbb{T}}$, see [21, Chapter 5]. Here, as before, $*$ denotes transposition. Note that the adjoint system is defined in the backward time (and is well defined only in the backward time). As follows from [30, Theorem 1], the Cauchy problem with the initial condition $x(t_0) = x_0 \in \mathbb{R}^d$ for (3.58) has a unique solution on $[t_0, t_1]_{\mathbb{T}}$. Let us denote by $\Omega(t, \tau)$ $(t \geq \tau, \tau, t \in [t_0, t_1]_{\mathbb{T}})$ the fundamental matrix of solutions of the system (3.58), normalized by the condition $\Omega(\tau, \tau) = I$, where I is the identity matrix. Then $\Omega(t, \tau) = e_A(t, \tau)$ is the matrix exponential function in the terminology of [20]. We also note that since we consider the solution of the system (3.58) only when $t \geq \tau$, the regressivity of A (by [30, Theorem 1]) is not needed. In what follows, for $\Omega(t, \tau)$, we will use the term 'matrizant'. From [30, Theorem 4], it also follows that the Cauchy problem with the initial condition $y(t_1) = y_0 \in \mathbb{R}^d$ for (3.59) has a unique solution on $[t_0, t_1]_{\mathbb{T}}$. Let us denote by $\Omega_*(t, \tau)$ $(t \leq \tau, \tau, t \in [t_0, t_1]_{\mathbb{T}})$ a matrizant of the system (3.59). We have the following result.

Lemma 3.6: *Let x and y be arbitrary solutions of systems (3.58) and (3.59), respectively, on $[t_0, t_1]_{\mathbb{T}}$. Then when $t \geq \tau$, $\tau, t \in [t_0, t_1]_{\mathbb{T}}$, we have*

$$\Omega^*(t, \tau)y(t) = y(\tau) \quad \text{and} \quad \Omega_*^*(\tau, t)x(\tau) = x(t). \tag{3.60}$$

***Proof*:** From [29, Formula (2)], we have

$$\begin{aligned}(x, y)^{\Delta}(t) &= (x^{\Delta}(t), y(\sigma(t))) + (x(t), y^{\Delta}(t)) \\ &= (A(t)x(t), y(\sigma(t))) + (x(t), -A^*(t)y(\sigma(t))) \\ &= 0.\end{aligned}$$

Therefore, for any solutions (3.58) and (3.59), we obtain

$$(x(t), y(t)) \equiv \text{constant}.$$

Relations (3.60) now follow from the fact that the columns of the matrizants $\Omega(t,\tau)$ and $\Omega_*(\tau,t)$ consist of the solutions of (3.58) and (3.59), respectively. □

3.4. Proof of Theorem 2.11

By the first statement of Lemma 3.2, the solution $x(t,\bar{\alpha},\bar{\gamma},x_0) = x(t)$ for $t \in [t_0,t_1]_{\mathbb{T}}$ lies in the domain D if $0 \le \alpha_i \le \varepsilon_0$, $0 \le \gamma_k < \varepsilon$, $|x_0 - \hat{x}_0| < \varepsilon_0$ for all $t \in [t_0,t_1]_{\mathbb{T}}$ when ε_0 is sufficiently small. Therefore, the pair $(x(t), u(t,\bar{\alpha},\bar{\tau},\bar{\nu},\bar{\gamma},\bar{r},\bar{z}))$, where u is defined by (3.4), is a controlled process of the problem (2.3)–(2.5).

We introduce the functions

$$I_i(\bar{\alpha},\bar{\gamma},x_0) = \Phi_i(x(t_0,\bar{\alpha},\bar{\gamma},x_0), x(t_1,\bar{\alpha},\bar{\gamma},x_0)),\ i \in \{0,\dots,n\}.$$

In view of the continuous differentiability of Φ_i in their arguments and the third statement of Lemma 3.2, the functions I_i are differentiable at the point $(\bar{0},\bar{0},\hat{x}_0)$. Now we can consider a finite-dimensional extremal problem (3.57) which is smooth in a cone:

$$\begin{aligned} I_0(\bar{\alpha},\bar{\gamma},x_0) &\to \inf, \\ I_i(\bar{\alpha},\bar{\gamma},x_0) &= 0, \quad i \in \{1,\dots,k\}, \\ I_i(\bar{\alpha},\bar{\gamma},x_0) &\le 0, \quad i \in \{k+1,\dots,n\}, \\ \alpha_m &\ge 0, \quad m \in \{1,\dots,N\}, \\ \gamma_j &\ge 0, \quad j \in \{1,\dots,\nu\}, \end{aligned} \tag{3.61}$$

for which the point $(\bar{0},\bar{0},\hat{x}_0)$ is a local solution and to which we can apply Theorem 3.5. By virtue of Theorem 3.5, let us apply the Lagrange multiplier rule to (3.61).

There exist Lagrange multipliers

$$\lambda = (\lambda_0,\lambda_1,\dots,\lambda_n),\ \mu = (\mu_1,\dots,\mu_N),\ \beta = (\beta_1,\dots,\beta_\nu),$$

nonzero simultaneously, such that for the Lagrange function

$$\Lambda(\bar{\alpha},\bar{\gamma},x_0,\lambda,\mu,\beta) = \sum_{i=1}^{n} \lambda_i I_i(\bar{\alpha},\bar{\gamma},x_0) - \sum_{i=1}^{N} \mu_i\alpha_i - \sum_{i=1}^{\nu} \beta_i\gamma_i, \tag{3.62}$$

the following conditions are satisfied:

(1) $\lambda_i \ge 0$, $i = 0$, $i \in \{k+1,\dots,n\}$; $\mu_i \ge 0$, $i \in \{1,\dots,N\}$; $\beta_i \ge 0$, $i \in \{1,\dots,\nu\}$;
(2) $\lambda_i\Phi_i(\hat{x}(t_0),\hat{x}(t_1)) = 0$, $i \in \{k+1,\dots,n\}$;
(3) $\hat{\Lambda}_{x_0} = 0$, where $\hat{\Lambda}_{x_0}$ is the partial derivative with respect to x_0;
(4) $\hat{\Lambda}_{\alpha_i} \ge 0$, $i \in \{1,\dots,N\}$; $\hat{\Lambda}_{\gamma_i} \ge 0$, $i \in \{1,\dots,\nu\}$, where

$$\begin{aligned} \hat{\Lambda}_{x_0} &= \Lambda_{x_0}(\bar{0},\bar{0},\hat{x}_0,\lambda,\mu,\beta), \\ \hat{\Lambda}_{\alpha_i} &= \Lambda_{\alpha_i}(\bar{0},\bar{0},\hat{x}_0,\lambda,\mu,\beta), \\ \hat{\Lambda}_{\gamma_i} &= \Lambda_{\gamma_i}(\bar{0},\bar{0},\hat{x}_0,\lambda,\mu,\beta). \end{aligned}$$

Conditions (1) and (2) indicate the fulfillment of conditions (1) and (2) of Theorem 2.11.

Now we derive conditions (3) and (4) using the formula of differentiation and superposition of Lemma 3.2. We have

$$\hat{\Lambda}_{x_0} = \Lambda_{x_0}(\bar{0}, \bar{0}, x_0, \lambda, \mu, \beta)\Big|_{x_0=\hat{x}_0} = L^*_{x_0} + L^*_{x_1} \frac{\partial x(t_1, \bar{0}, \bar{0}, x_0)}{\partial x_0}\Big|_{x_0=\hat{x}_0} = 0. \tag{3.63}$$

Here, $L = \sum_{i=0}^{n} \lambda_i \Phi_i(x(t_0), x(t_1))$ is the Lagrange function of the problem (2.3)–(2.5). Relation (3.63), considering (3.8), takes the form

$$L^*_{x_0} + L^*_{x_1} \Omega(t_1, t_0) = 0. \tag{3.64}$$

We denote by Ψ the solution of the adjoint system (2.9) with the initial condition

$$\Psi(t_1) = -L_{x_1}. \tag{3.65}$$

Then (3.64) takes the form

$$L^*_{x_0} - \Psi^*(t_1)\Omega(t_1, t_0) = 0.$$

We use Lemma 3.6. Inserting the first relation from (3.60) for $t = t_1$ and $\tau = t_0$, the last relation takes the form $L^*_{x_0} - \Psi(t_0) = 0$, which together with (3.65) implies realization of the transversality conditions (3) in Theorem 2.11.

We now consider the condition (4):

$$\hat{\Lambda}_{\alpha_i} = \Lambda_{\alpha_i} = \left(L_{x_1}, \frac{\partial x(t_1, \alpha_i, \bar{0}, \hat{x}_0)}{\partial \alpha_i}\Big|_{\alpha_i=0} \right) - \mu_i \geq 0.$$

The last relation, in view of (3.9), takes the form

$$\big(L_{x_1}, \Omega(t_1, \tau_i)\Delta f(\tau_i)\big) - \mu_i \geq 0, \tag{3.66}$$

where $\Delta f(\tau_i) = f(\tau_i, \hat{x}(\tau_i), v_i) - f(\tau_i, \hat{x}(\tau_i), \hat{u}(\tau_i))$. Then we have

$$\big(\Omega^*(t_1, \tau_i)L_{x_1}, \Delta f(\tau_i)\big) \geq \mu_i, \ i \in \{1, \dots, N\}. \tag{3.67}$$

Inequality (3.67) implies that if all $\lambda_i = 0$, that is, the Lagrange function L of the problem (2.3)–(2.5) is identically equal to zero, then $\mu_i = 0$, $i \in \{1, \dots, N\}$.

Considering now the first relation (3.60) when $t = t_1$, $\tau = \tau_i$ and (3.65), (3.67) implies $\big(\Psi(\tau_i), \Delta f(\tau_i)\big) \leq 0$. That in terms of the Pontryagin function (Definition 2.10) means the fulfillment of condition (4)(b) of Theorem 2.11 at the point τ_i.

Similarly, in view of (3.10),

$$\hat{\Lambda}_{\gamma_i} = \left(L_{x_1}, \Omega(t_1, \sigma(r_i))\mu(r_i) \frac{\partial f(r_i, \hat{x}(r_i), \hat{u}(r_i))}{\partial u} \big(z_i - \hat{u}(r_i)\big) \right) - \beta_i \geq 0. \tag{3.68}$$

Again, if all $\lambda_i = 0$, then all $\beta_i = 0$, $i \in \{1, \dots, \nu\}$. Thus not all λ_i equal zero. From (3.68), we now have

$$\mu(r_i) \left(\Psi^\sigma(r_i), \frac{\partial f(r_i, \hat{x}(r_i), \hat{u}(r_i))}{\partial u} \big(z_i - \hat{u}(r_i)\big) \right) \leq 0,$$

which means fulfillment of condition (4)(a) of Theorem 2.11 at the point r_i.

These Lagrange multipliers $\lambda = (\lambda_1, \dots, \lambda_n)$ and, therefore, the vector Ψ, depend on the 'package of needle variations'. The existence of Lagrange multipliers $\lambda = (\lambda_1, \dots, \lambda_n)$ that are 'universal', i.e., independent of the 'package of needle variations', is proved similarly as in the classical case ($\mathbb{T} = \mathbb{R}^1$) using a well-known lemma of a centered system of compact sets.

The proof of Theorem 2.11 is complete.

3.5. Proof of Theorem 2.13

Since the functions Φ_i in (2.12) and (2.14) are convex, the function $L = \sum_{i=1}^{n} \lambda_i \Phi_i(x(t_0), x(t_1))$ of the problem (2.12)–(2.14) is convex, too. Let (x, u) be an arbitrary controlled admissible process and $(\hat{x}, \hat{u})$ be the process which appears in Theorem 2.13. We take an arbitrary $\varepsilon > 0$. Then we have

$$\begin{aligned} &L(\lambda, \varepsilon x(t_0) + (1-\varepsilon)\hat{x}(t_0), \varepsilon x(t_1) + (1-\varepsilon)\hat{x}(t_1)) \\ &\quad \le \varepsilon L(\lambda, x(t_0), x(t_1)) + (1-\varepsilon) L(\lambda, \hat{x}(t_0), \hat{x}(t_1)). \end{aligned}$$

Hence

$$\begin{aligned} &L\left(\lambda, \hat{x}(t_0) + \varepsilon\left(x(t_0) - \hat{x}(t_0)\right), \hat{x}(t_1) + \varepsilon\left(x(t_1) - \hat{x}(t_1)\right)\right) \\ &\quad \le L(\lambda, \hat{x}(t_0), \hat{x}(t_1)) + \varepsilon\left(L(\lambda, x(t_0), x(t_1)) - L(\lambda, \hat{x}(t_0), \hat{x}(t_1))\right), \end{aligned}$$

i.e.,

$$\begin{aligned} &\frac{L\left(\lambda, \hat{x}(t_0) + \varepsilon\Delta\hat{x}(t_0), \hat{x}(t_1) + \varepsilon\Delta\hat{x}(t_1)\right) - L(\lambda, \hat{x}(t_0), \hat{x}(t_1))}{\varepsilon} \\ &\qquad \le L(\lambda, x(t_0), x(t_1)) - L(\lambda, \hat{x}(t_0), \hat{x}(t_1)), \end{aligned} \tag{3.69}$$

where we denote $\Delta\hat{x}(t_0) = x(t_0) - \hat{x}(t_0)$ and $\Delta\hat{x}(t_1) = x(t_1) - \hat{x}(t_1)$.

By passing to the limit for $\varepsilon \to 0$ in (3.69) and using the differentiability of L, we obtain

$$\left(L_{x_0}, \Delta\hat{x}(t_0)\right) + \left(L_{x_1}, \Delta\hat{x}(t_1)\right) \le L(\lambda, x(t_0), x(t_1)) - L(\lambda, \hat{x}(t_0), \hat{x}(t_1)). \tag{3.70}$$

Let $\Omega(t, t_0)$ be a matrizant of the system $x^\Delta = A(t)x$ at the point t_0. Since $\hat{x}$ is a solution for the system $x^\Delta = A(t)x + B(t)\hat{u}(t)$ and x is a solution for the system $x^\Delta = A(t)x + B(t)u(t)$, $\Delta\hat{x} = x - \hat{x}$ satisfies $\Delta\hat{x}^\Delta = A(t)\Delta\hat{x} + B(t)(u(t) - \hat{u}(t))$. Using [20, Theorem 5.24], we have that $\Delta x(t)$ for $t \ge t_0$, $t \in [t_0, t_1]_\mathbb{T}$, has the form

$$\Delta\hat{x}(t) = \Omega(t, t_0)\Delta\hat{x}(t_0) + \int_{[t_0, t)_\mathbb{T}} \Omega(t, \sigma(s))B(s)(u(s) - \hat{u}(s))\Delta s. \tag{3.71}$$

From (3.71), when $t = t_1$, we have

$$\Delta\hat{x}(t_1) = \Omega(t_1, t_0)\Delta\hat{x}(t_0) + \int_{[t_0, t_1)_\mathbb{T}} \Omega(t_1, \sigma(s))B(s)(u(s) - \hat{u}(s))\Delta s. \tag{3.72}$$

According to condition (3) of Theorem 2.13,

$$L_{x_0} = \Psi(t_0) = 0, \quad \text{and hence} \quad \big(L_{x_0} - \Psi(t_0),\ \Delta\hat{x}(t_0)\big) = 0.$$

Taking into account (3.60) and (3.65), the latter relation takes the form

$$\big(L_{x_0} + \Omega^*(t_1, t_0)L_{x_1}, \Delta\hat{x}(t_0)\big) = 0,$$

i.e.,

$$(L_{x_0}, \Delta\hat{x}(t_0)) + \big(L_{x_1}, \Omega(t_1, t_0)\Delta\hat{x}(t_0)\big) = 0. \tag{3.73}$$

Since the condition of maximum (4) of Theorem 2.13 is satisfied, we obtain

$$(\Psi^\sigma(s), B(s)u(s)) \le (\Psi^\sigma(s), B(s)\hat{u}(s))$$

and, therefore

$$-\int_{[t_0,t_1)_{\mathbb{T}}} (\Psi^\sigma(s), B(s)(u(s) - \hat{u}(s)))\Delta s \ge 0.$$

By (3.60), we have

$$\int_{[t_0,t_1)_{\mathbb{T}}} (\Omega^*(t_1, \sigma(s))L_{x_1}, B(s)(u(s) - \hat{u}(s)))\Delta s \ge 0.$$

Hence

$$\left(L_{x_1}, \int_{[t_0,t_1)_{\mathbb{T}}} \Omega(t_1, \sigma(s))B(s)(u(s) - \hat{u}(s))\Delta s\right) \ge 0. \tag{3.74}$$

Now summing (3.73) and (3.74), we get

$$\begin{aligned}&(L_{x_0}, \Delta\hat{x}(t_0))\\ &\quad + \left(L_{x_1}, \Omega(t_1, t_0)\Delta\hat{x}(t_0) + \int_{[t_0,t_1)_{\mathbb{T}}} \Omega(t_1, \sigma(s))B(s)(u(s) - \hat{u}(s))\Delta s\right) \ge 0.\end{aligned}$$

Therefore

$$(L_{x_0}, \Delta\hat{x}(t_0)) + (L_{x_1}, \Delta\hat{x}(t_1)) \ge 0.$$

Now taking into account (3.70), we get

$$L(\lambda, x(t_0), x(t_1)) \ge L(\lambda, \hat{x}(t_0), \hat{x}(t_1)),$$

i.e.,

$$\sum_{i=0}^{n} \lambda_i \Phi_i(x(t_0), x(t_1)) \ge \sum_{i=0}^{n} \lambda_i \Phi_i(\hat{x}(t_0), \hat{x}(t_1)).$$

Then

$$\begin{aligned}&\lambda_0\Phi_0(x(t_0), x(t_1)) + \sum_{i=1}^{n} \lambda_i \Phi_i(x(t_0), x(t_1))\\ &\quad \ge \lambda_0\Phi_0(\hat{x}(t_0), \hat{x}(t_1)) + \sum_{i=1}^{n} \lambda_i \Phi_i(\hat{x}(t_0), \hat{x}(t_1)),\end{aligned}$$

so that

$$\Phi_0(x(t_0), x(t_1)) \geq \Phi_0(\hat{x}(t_0), \hat{x}(t_1)).$$

Therefore, the process $(\hat{x}, \hat{u})$ is optimal.

The proof of Theorem 2.13 is complete.

Remark 3.7: Theorem 2.13 shows that the pair $(\hat{x}, \hat{u})$ yields a global minimum of (2.12)–(2.14) in the class of admissible processes.

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No potential conflict of interest was reported by the authors.

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